

Monday.

8

Time: 20 min

Surname:

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Initials:

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Student number:

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Group:

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M-number

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Use a pen to complete the test. Show all your steps. Use interval notation, where applicable. You are not allowed to use a calculator. The test is printed on both pages.

QUESTION 1 [2]

Can a function be continuous at $x = 2$ but not differentiable at $x = 2$? If your answer is "Yes", prove the statement, if your answer is "No", give a counterexample.

Yes. $f(x) = |x-2|$. f is continuous at $x=2$.

$$\lim_{x \rightarrow 2^-} \frac{f(x) - f(2)}{x-2} = \lim_{x \rightarrow 2^-} \frac{-(x-2) - 0}{x-2} = -1$$

$$\lim_{x \rightarrow 2^+} \frac{f(x) - f(2)}{x-2} = \lim_{x \rightarrow 2^+} \frac{(x-2) - 0}{x-2} = 1.$$

$\Rightarrow f'(2)$ does not exist $\Rightarrow f$ not differentiable at $x=2$

QUESTION 2 [2]

Use the definition of the derivative to find $f'(a)$, where $f(x) = x^2 + x$ and $a = 2$.

$$f'(2) = \lim_{x \rightarrow 2} \frac{f(x) - f(2)}{x-2}$$

$$= \lim_{x \rightarrow 2} \frac{(x^2 + x) - (2^2 + 2)}{x-2}$$

$$= \lim_{x \rightarrow 2} \frac{(x+3)(x-2)}{x-2}$$

$$= \lim_{x \rightarrow 2} (x+3) = 5.$$

$$\begin{aligned} x^2 + x - 6 \\ = (x+3)(x-2) \end{aligned}$$

QUESTION 3 [1, 1.5, 1.5]

Find the derivative of the following functions. Do not simplify your answer.

(i) $f(t) = 2^{3-t}$

$$f'(t) = 2^{3-t} \times \ln 2 \times \frac{d}{dt}(3-t)$$

$$= 2^{3-t} \ln 2 (-1)$$

(ii) $f(t) = \sin^2(7t) = (\sin 7t)^2$

$$f'(t) = 2(\sin 7t) \times (\cos 7t) \times 7.$$

(iii) $f(x) = \frac{x}{1+\cos x}.$

$$f'(x) = \frac{\left(\frac{d}{dx} x\right)(1+\cos x) - x \frac{d}{dx}(1+\cos x)}{(1+\cos x)^2}$$

$$= \frac{1+\cos x - x(-\sin x)}{(1+\cos x)^2}$$